

# YIXUAN HE

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Assistant Professor @ Arizona State University

## EDUCATION

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**University of Oxford, OU**

Oct. 2020 - Jun. 2024

*D.Phil. (PhD) student in Statistics*

*Supervisors: Mihai Cucuringu and Gesine Reinert*

- **Clarendon Scholar (fully funded)**, Balliol Jason Hu Scholarship
- Thesis title: **Graph Neural Networks for Network Analysis**

**University of Edinburgh, UoE**

Sept. 2018 - Jun. 2020

*BSc (Hons) in Math and Stats*

*Supervisor: Timothy I. Cannings*

- GPA: **92.33%** Avg., **Rank # 1** in Math & Stats, **First Class**

**University of California, Berkeley**

Jan. 2018 - May 2018

*Semester Exchange*

- GPA: **4.0/4.0**

**South China University of Technology, SCUT**

Sept. 2016 - Jun. 2020

*BSc Mathematics and Applied Mathematics*

*Supervisor: Delu Zeng*

- Rank: **1/60** (Outstanding graduate)
- Overall GPA for 2016 - 18: **3.99/4.0**, Major GPA: **4.0/4.0**; transferred to UoE on 2+2 program
- Winning 2 years of **National Scholarship** granted by Ministry of Education of China
- **650/710** in CET6, **top 0.02 % of students** in SCUT

## RESEARCH EXPERIENCE

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***OU: Graph Neural Networks for Network Analysis***

Oct. 2020 - Jun. 2024

*Doctoral Research*

*Supervisors: Gesine Reinert, Mihai Cucuringu*

- Worked on signed, directed, and temporal GNNs, as well as GNNs for signal recovery.
- Developed open-source libraries: creator of PyG Signed Directed & contributor of PyG Temporal.
- Collaborated with Amazon as an applied scientist intern (Mentor: David Wipf).

***NetEase Games: Behaviour-based Anti-cheating in FPS Games***

Jun. 2020 - Aug. 2020

*Summer Internship*

*Supervisor: Xin Wen*

- Extracted features from logs using multiprocessing & built multi-input single-output models with TensorFlow 2.2.

***UoE: AUROC vs AUPRC: A Comparison of Performance Measures in Classification Problems***

Sept. 2019 - Apr. 2020

*UoE Honors Project*

*Supervisor: Timothy I. Cannings*

- Collected data, compared measure scores, **analyzed measure properties** & data properties.

***UCLA CSST summer research/REU: Deforestation Brazil***

Jun. 2019 - Sept. 2019

*Funded by NGA & NSF*

*Supervisors: Christian Parkinson, Andrea L. Bertozzi, Stanley J. Osher*

- Conducted data mining, data analysis, matlab programming & model construction.
- Implemented **level set method**, numerical PDEs, optimal path searching, satellite image processing, Maximum Penalized Likelihood Estimation (MPLE), time series simulation etc.

- **Delivered a talk** at IPAM’s “The Level Set Collective” seminar invited by Professor Stanley Osher.
- **Presented a poster** at Intelligence Community Academic Research Symposium (ICARS) 2019.

***SCUT: Robust Image Salient Object Perception Prior for Image Segmentation based on PDEs Related Methods***

May 2018 - May 2020

*Funded by National Natural Science Foundation of China (NSFC)*

*Supervisor: Delu Zeng*

- Studied level set methods and **neural networks** on **semantic image segmentation**.
- Worked on lung nodule segmentation using MEnet.
- Utilized Linux, Colab, Caffe, TensorFlow, PyTorch & programming in Python.

***SCUT: Efficiency of Collaborative Innovation Knowledge Output in Professional Towns in Internet + Times***

May 2017 - May 2018

*Funded by MOE of PRC*

*Supervisors: Shifu Wang, Miaoqi Zhao, Shenquan Liu*

- **Cross-disciplinary:** Math & Urban Planning.
- Responsible: data mining & analysis, paper writing.
- **Led** the project as part of the National Training Program of Innovation for Undergraduates.

## MAJOR AWARDS

|                        |                                                                                             |
|------------------------|---------------------------------------------------------------------------------------------|
| <b>Grant</b>           | 2026 NVIDIA Academic Grant Program Award                                                    |
| <b>Grant&amp;Honor</b> | 2026 Jetstream2 NAIRR AI Fellows Program                                                    |
| <b>Grant</b>           | 2024 G-Research June Grant (mainly for attending ICML 2024)                                 |
| <b>Grant</b>           | 2024 ICML Official Financial Aid                                                            |
| <b>Grant</b>           | 2024 ICLR Official Financial Assistance                                                     |
| <b>Award</b>           | 2023/2024 Academic Year Oxford University Blues Award for Sports Excellence                 |
| <b>Grant</b>           | 2024 Balliol College Support Grant                                                          |
| <b>Grant</b>           | 2023-24 Balliol College Floreat Grant for Sports                                            |
| <b>Grant</b>           | 2022-23 Balliol College Floreat Grant for Sports                                            |
| <b>Grant</b>           | 2022 ICML Official Participant Award                                                        |
| <b>Grant</b>           | 2022 Simons Institute Summer School Fund                                                    |
| <b>Grant</b>           | 2022 Balliol’s fund on IMA Conf. on The Mathematical Challenges of Big Data reg.            |
| <b>Award</b>           | 2022 SIAM Student Travel Award for SDM’22                                                   |
| <b>Award</b>           | 2022 SIGIR Student Registration Award for WSDM’22                                           |
| <b>Grant</b>           | 2021 Balliol’s fund on Complex Networks 2021 reg.                                           |
| <b>Grant</b>           | 2021 Balliol’s fund on NeurIPS 2021 reg.                                                    |
| <b>Grant</b>           | 2021 Balliol’s fund on ACM membership & CIKM’21 registration                                |
| <b>Prize</b>           | 2020 Napier Medal and Gangadhar Balwant Gadgil Prize ( <b>top</b> Math student @UoE)        |
| <b>Prize</b>           | 2020 Lawley Memorial Prize ( <b>top</b> student in Math & Stats @UoE)                       |
| <b>Scholarship</b>     | 2020 <b>Clarendon Scholarship</b> & Balliol Jason Hu scholarship at Oxford                  |
| <b>Honor</b>           | 2020 Outstanding Graduate at SCUT Math Dept.                                                |
| <b>Prize</b>           | 2019 Arthur Erdelyi Prize (For top three performing Math students @UoE)                     |
| <b>Prize</b>           | 2019 James Ward Memorial Prize ( <b>top</b> student in Math & Stats @UoE)                   |
| <b>Second Prize</b>    | 2018 ICM/MCM International <b>Mathematical Modeling</b> Competition                         |
| <b>Grand Prize</b>     | 2018 Math Competition of SCUT (Math Group) ( <b>Rank # 1</b> )                              |
| <b>Honor</b>           | 2017-2018 one of the 10 <b>Role Models to Merit Students</b> at SCUT                        |
| <b>Prize</b>           | 2017 & 2018 <b>National Scholarship</b>                                                     |
| <b>Second Prize</b>    | 2017 “9 <sup>th</sup> National College Student <b>Math Competition</b> (Math)” in Guangdong |
| <b>Second Prize</b>    | 2017 National College Student Mathematical Modeling Competition (Guangdong)                 |

## PUBLICATIONS

\* indicates equal contribution. All non-arXiv publications are peer-reviewed. "Talk" means additional oral presentations.

- Sandel, A., **He, Y.**, ..., & Mitani, J. Lethal Conflict Following Group Fission in Wild Chimpanzees. (**Science**, 2026)
- Liang, P., Yang, C., **He, Y.**, Pan, R., & Chen, Y. Topology-Guided Graph Pre-training and Prompt Learning on Directed Graphs. (**TMLR**, 2026)
- Yilmaz, N., Patel, M., Kusumba, A., **He, Y.**, & Yang, Y. MentalBlackboard: Evaluating Spatial Visualization via Mathematical Transformations. (arXiv 2026)
- Ockerman, S., Gueroudji, A., Mallick, T., **He, Y.**, Pouchard, L., Ross, R., & Venkataraman, S. PGT-I: Scaling Spatiotemporal GNNs with Memory-Efficient Distributed Training. (**SC25**)
- **He, Y.**, Sandel A., Wipf D., Cucuringu M., Mitani J. & Reinert G. Learning to Fuse Temporal Proximity Networks: A Case Study in Chimpanzee Social Interactions. (arXiv 2025)
- Aminian, G.\*, **He, Y.\***, Reinert, G., Szpruch, L., Cohen, SN. Generalization Error of Graph Neural Networks in the Mean-field Regime. (**ICML 2024**)
- **He, Y.**, Reinert, G., Wipf, D., & Cucuringu, M. Robust Angular Synchronization via Directed Graph Neural Networks. (**ICLR 2024**)
- Steach, H., Viswanath, S., **He, Y.**, Zhang, X., Ivanova, N., Hirn, M., Perlmutter, M., & Krishnaswamy, S. Inferring Metabolic States from Single Cell Transcriptomic Data via Geometric Deep Learning. (**RECOMB 2024**)
- **He, Y.**, Zhang, X., Huang, J., Rozemberczki, B., Cucuringu, M., & Reinert, G. PyTorch Geometric Signed Directed: A Software Package on Graph Neural Networks for Signed and Directed Graphs. (**LoG 2023, Open-Source Software with 100+ stars**)
- Geng, H., Chen, C., **He, Y.**, Zeng, G., Han, Z., Chai, H., Yan, J. Pyramid Graph Neural Network: A Graph Sampling and Filtering Approach for Multi-scale Disentangled Representations. (**KDD 2023, Talk**)
- Wu, Q., Yang, C., Zhao, W., **He, Y.**, Wipf, D., & Yan, J. DIFFormer: Scalable (Graph) Transformers Induced by Energy Constrained Diffusion. (**ICLR 2023, Spotlight Talk**)
- **He, Y.**, Reinert, G., & Cucuringu, M. DIGRAC: Digraph Clustering Based on Flow Imbalance. (**LoG 2022**)
- **He, Y.**, Perlmutter, M., Reinert, G., & Cucuringu, M. MSGNN: A Spectral Graph Neural Network Based on a Novel Magnetic Signed Laplacian. (**LoG 2022**)
- Wang, X., Chen, S., **He, Y.**, Wang, M., Gan, Q., & Yan, J. CEP3: Community Event Prediction with Neural Point Process on Graph. (**LoG 2022**)
- **He, Y.**, Gan, Q., Wipf, D., Reinert, G. D., Yan, J., & Cucuringu, M. GNNRank: Learning Global Rankings from Pairwise Comparisons via Directed Graph Neural Networks. (**ICML 2022, Talk**)
- **He, Y.**, Reinert, G., Wang, S., & Cucuringu, M. SSSNET: Semi-Supervised Signed Network Clustering. (**SDM 2022, Talk**)
- **He, Y.** GNNs for Node Clustering in Signed and Directed Networks. (**WSDM-DC 2022, Talk**)
- Zhang, X., **He, Y.**, Brugnone, N., Perlmutter, M., & Hirn, M. MagNet: A Neural Network for Directed Graphs. (**NeurIPS 2021**)
- Rozemberczki, B., Scherer, P., **He, Y.**, Panagopoulos, G., Riedel, A., Astefanoaei, M., ... & Sarkar, R. PyTorch Geometric Temporal: Spatiotemporal Signal Processing with Neural Machine Learning Models. (**CIKM 2021 best paper award, Open-Source Software with 2.9k stars, 400+ forks**)
- **He, Y.**, Hu, T., & Zeng, D. Scan-flood Fill (SCAFF): An Efficient Automatic Precise Region Filling Algorithm for Complicated Regions. (**CVPR 2019 workshop**)
- Zeng, D., **He, Y.**, Liu, L., Chen, Z., Huang, J., Chen, J., & Paisley, J. Ro-SOS: Metric Expression Network (MEnet) for Robust Salient Object Segmentation. (arXiv 2018)

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**TEACHING EXPERIENCE**

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**Arizona State University (ASU) Assistant Professor**

Sept. 2024 -

- Grad course: Statistics for Bio Data Sci II (F24)
- Undergrad courses: Regression and Time Series Analyses (F25, F26), Stat Analysis for Researchers (SP25, SP26), Nonparametric Statistics (SP25, F26), Statistical Computing (SP26)

**University of Oxford Teaching Assistant**

Dec. 2020 - Dec. 2022

- Courses to grade and discuss a problem in the problem classes: Applied Statistics (F22), Advanced Topics in Statistical Machine Learning (SP22), Computational Statistics (SP21)
- Course to demonstrate and answer questions: Statistics and Data management (for DTC in Dept. of Zoology, F20)

**UoE's Digital Skills Trainer - Python**

Dec. 2019 - May 2020

- **Delivered Python training courses**, reviewed & improved existing Python training.
- Recorded lectures: Introduction to Python and Python for Data Science.

**Teaching Support in a Primary School in Longsheng Rural Village**

July 2017

- Devised & delivered lectures in singing, dancing, craftsmanship & story-telling.
- **Leader** in hosting the closing ceremony.
- **Visited families** to collect information & offer suggestions.

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**ACADEMIC SERVICES**

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**Reviewer/Area Chair**

- Conferences: GFM @ ICML 2026 (AC), NeurIPS 2026, ICML 2026, ICLR 2026, AISTATS 2026, AAAI 2026, NeurIPS 2025, ICML 2025, AISTATS 2025, ICLR 2025, AAAI 2025, NeurIPS 2024, ICML 2024, ICLR 2024, AAAI 2024, NeurIPS 2023, ICML 2023, WWW 2023, AAAI 2023, NeurIPS 2022, SIGKDD 2022, ICML 2022, WSDM 2022
- Journals: IEEE Transactions on Signal and Information Processing over Networks, IEEE Transactions on Neural Networks and Learning Systems (TNNLS), IEEE Transactions on Knowledge and Data Engineering (TKDE), Journal of Complex Networks, Journal of Quality Technology, Statistics and Computing

**(Co-)Organizer**

- ASU Machine Learning Day 2026, ASU
- Learning on Graphs Conference 2025, ASU (local host)
- ASU Machine Learning Day 2025, ASU
- AI Seminar @ ASU, ASU
- Learning on Graphs Local Meetup @ Arizona 2024, ASU
- LoG<sup>2</sup> @ Oxford: the Learning on Graphs and Geometry seminar series, University of Oxford

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**WORK EXPERIENCE**

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**Assistant Professor @ Arizona State University**

Sept. 2024 -

- Teach Statistics @ School of Mathematical and Natural Sciences.
- Conduct research & mentor students.
- Organize & participate in research activities.

**Applied Scientist Intern @ Amazon Web Services**

Nov. 2021 - Aug. 2024

- Mentor: David Wipf.
- Conducted research on graph neural networks.

**Game Development Engineer @ NetEase Games**

Jun. 2020 - Aug. 2020

- Project: Behaviour-based Anti-cheating in FPS Games (Supervisor: Xin Wen)
- Selected as **Outstanding Intern** to present my work.

**Student Assistant for School of Mathematics @ SCUT**

Sept. 2016 - Aug. 2017

- Assisted in organizing materials & releasing notifications.
- Outstanding individual in work-study of South China University of Technology.

**CONFERENCES/SUMMER SCHOOLS**

|                                                                           |                        |            |
|---------------------------------------------------------------------------|------------------------|------------|
| 2026 AAAS Annual Meeting – Science @ Scale                                | Volunteer              | Feb. 2026  |
| 4th Learning on Graphs (LoG) Conference                                   | Organizer & Local Host | Dec. 2025  |
| 41st International Conference on Machine Learning ( <b>ICML</b> )         | Author                 | Jul. 2024  |
| CNRS GdR IASIS Graph Learning Day                                         | Speaker                | Jun. 2024  |
| 12th International Conference on Learning Representations ( <b>ICLR</b> ) | Author                 | May 2024   |
| 2nd Learning on Graphs (LoG) Conference                                   | Author                 | Nov. 2023  |
| Workshop: statistical learning on LARge scale GRaphs (LARGR)              | Presenter              | Mar. 2023  |
| 1st Learning on Graphs (LoG) Conference                                   | Author                 | Dec. 2022  |
| 4th IMA Conference on The Mathematical Challenges of Big Data             | Speaker                | Sept. 2022 |
| 39th International Conference on Machine Learning ( <b>ICML</b> )         | Author                 | Jul. 2022  |
| SIAM International Conference on Data Mining ( <b>SDM</b> )               | Author                 | Apr. 2022  |
| 15th ACM Int. Conf. on Web Search and Data Mining ( <b>WSDM</b> )         | Doctoral Consortium    | Feb.2022   |
| 35th Conf. on Neural Information Processing Systems ( <b>NeurIPS</b> )    | Author                 | Dec.2021   |
| The 10th Int. Conf. on Complex Networks and their Applications            | Author                 | Nov.2021   |
| Int. Conf. on Information and Knowledge Management ( <b>CIKM</b> )        | Author                 | Nov.2021   |
| Intelligence Community Academic Research Symposium ( <b>ICARS</b> )       | Poster Presenter       | Sept. 2019 |
| Computer Vision and Pattern Recognition ( <b>CVPR</b> )                   | Author                 | Jun. 2019  |

**COMPUTER SKILLS**

|                       |                                                                            |
|-----------------------|----------------------------------------------------------------------------|
| Programming:          | <b>Python (Advanced)</b> , C++ (Intermediate), VB (Basic) & Pascal (Basic) |
| Statistical Analysis: | <b>R (Advanced)</b> , SQL (Intermediate) & SPSS (Intermediate)             |
| Modelling:            | <b>Matlab (Advanced)</b> , Maple (Intermediate) & Xpress (Intermediate)    |
| Paper writing:        | <b>Latex (Advanced)</b>                                                    |

**POSITIONS OF RESPONSIBILITY****President of Oxford University Table Tennis Club (OUTTC)**

May 2023 - May 2024

- **Organized** club activities, promoted the club, organized training, booked & coordinated facilities.
- Participated in table tennis **competitions** as a squad member.
- **Initialized** the use of an Oxford mailing list to send notifications (600+ subscribers) & **managed** a club Facebook group (1.2k members).

**Director of Dept. of Visiting Scholars in Oxford Chinese Students & Scholars Association (OXCSSA)**

July 2021 - July 2022

- **Co-organized** 2021 welcoming event for new students to University of Oxford.

- **Organized** events for visiting scholars: visiting scholars' tea party, hiking, formal, punting etc.
- Collaborated with other departments to hold virtual and in-person talks.

**Head of Class**

Sept. 2013 - Jun. 2018

- Conveyed notifications, organized activities & led by example.
- 2017 & 2018 South China University of Technology **Outstanding Student Leader**.
- 2016-2017 **outstanding class** of SCUT & 2017-2018 top-16 classes of SCUT.

**University Student Representative**

May 2017 - May 2018

- Actively listened to students' opinions & participated in raising proposals.
- Outstanding Student Representative of South China University of Technology in 2017.

**Member of Public Relationship Dept., Math Student Union**

Sept. 2016 - Oct. 2017

- **Led** the hosting of a social party in School of Mathematics.
- 2017 **Star** in Mathematics Student Union.